



CL-2001 Data Structures Lab # 12

Objectives:

- Graph
- Adjacency Matrix
- Adjacency List

Note: Carefully read the following instructions (*Each instruction contains a weightage*)

1. There must be a block of comments at start of every question's code by students; the block should contain brief description about functionality of code.
2. Comment on every function about its functionality.
3. Use understandable name of variables.
4. Proper indentation of code is essential
5. Write a C++ statement(s) for each of the following task one after the other, in the same order.
6. Make a Microsoft Word file and paste all of your C++ code with all possible screenshots of every **task output in MS word and submit .cpp file with word file.**
7. Make separate .cpp files for all tasks and use this format **22F-1234_Task1.cpp.**
8. First think about statement problems and then write/draw your logic on copy.
9. After copy pencil work, code the problem statement on MS Studio C++ compiler.
10. At the end when you done your tasks, attached C++ created files in MS word file and make your submission on Google classroom. (Make sure your submission is completed).
11. Please submit your file in this format **22F-1234_L1.**
12. Do not submit your assignment **after the deadline.**
- 13. Do not copy code from any source otherwise you will be penalized with negative marks.**



Problem 1 | Implement Adjacency Matrix

Given a undirected Graph of N vertices 1 to N and M edges in form of 2D array. Array `[][]` whose every row consists of two numbers X and Y which denotes that there is an edge between X and Y, the task is to write c++ program to create Adjacency Matrix of the given Graph.

The following list of edges are given to you by using this edge list show the adjacency matrix and count that how many edges each node has.

[0,1], [0,6], [0,8], [1,4], [1,6], [1,9], [2,4], [2,6], [3,4], [3,5], [3,8], [4,5], [4,9], [7,8], [7,9]

Problem 2 | Implement Adjacency List

Implement Adjacency list for undirected Graph of N vertices 1 to N and M edges. Test your code on at least 5 vertices.

Problem 3 | Implement Adjacency List/ Matrix(Your choice)

The transportation network consists of a set of cities connected by roads. Each road has a specific distance associated with it. The company's fleet comprises several vehicles with different capacities to carry packages. Each vehicle can start and end its route from the company's headquarters, which serves as the central hub for all deliveries.

Graph Representation: Create a C++ class to represent the transportation network as a weighted graph, where each city corresponds to a node, and each road between cities is an edge with its respective distance.

Vehicle Capacities: Input the capacities of each vehicle in the fleet. The capacities represent the maximum number of packages or the maximum weight the vehicle can carry.

Package Information: Gather information about the packages to be delivered, including their destinations, sizes, and weights.

Return to Headquarters: Ensure that each vehicle returns to the company's headquarters after completing its deliveries. Headquarters will be the starting and ending point for all routes.

Visualization: Implement a visualization module to display the planned routes and deliveries, allowing visual verification and assessment of the proposed solution.

Problem 4 | Implement BFS

Implement BFS using Adjacency Matrix.



Problem 5 | Implement DFS

Implement DFS using Adjacency list.

😊 Best of luck